

Towards Qualitative Measurement at Scale: A Prompt-Engineering Framework for Large-Scale Analysis of Deliberative Quality in Parliamentary Debates

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Abstract

This paper investigates whether Large Language Models (LLMs), guided by prompt engineering, can automate the complex Discourse Quality Index (DQI) for measuring deliberative quality. Evaluating state-of-the-art models from OpenAI and Google by varying In-Context Learning (ICL) examples (0-50) shows LLMs achieve high fidelity, comparable to human annotators based on standard reliability metrics. Performance improves markedly with few examples, plateauing around 25-50 examples. While both models perform well, differences highlight the interplay between model selection and ICL strategy. Error analysis identifies specific DQI dimensions requiring further improvement, suggesting future work on advanced reasoning prompts. This study confirms LLM viability for scaling DQI measurement and provides practical guidance on optimizing ICL strategies. Additionally, it contributes a modular, adaptable AI engineering pipeline that researchers can leverage for their own prompting experiments across various measurement tasks.

1 Introduction

The ability to measure complex linguistic, psychological, and social constructs from textual data is central to many research questions in the social sciences and humanities. In political science, for example, understanding the quality of deliberative communication in parliamentary debates is essential for evaluating the health of democratic institutions and the legitimacy of political decisions (Habermas, 1996; Bächtiger et al., 2018; Steenbergen et al., 2003b). Similarly, tracking the evolution of social norms and cultural beliefs over time involves measuring latent dimensions from large historical corpora (Garg et al., 2018; Grossmann and Varnum, 2015).

Traditionally, measuring such rich, high-dimensional constructs from text has relied on qualitative methods like expert human annotation. Approaches like the Discourse Quality Index (DQI; Steenbergen et al. 2003b) leverage the knowledge and interpretive abilities of trained coders to make judgments about multiple dimensions of textual data. While these methods can yield valid and reliable measurements, they are often prohibitively expensive and time-consuming to apply to large datasets. As a result, many researchers have turned to more scalable computational methods, such as dictionary-based analysis (Tausczik and Pennebaker, 2010), supervised machine learning (Rudkowsky et al., 2018), and unsupervised topic modeling (Blei et al., 2003). However, these automated methods typically measure relatively narrow constructs, as they often struggle to capture the deeper meaning and context embedded in text (Grimmer and Stewart, 2013).

Recent advances in Large Language Models (LLMs) offer a promising pathway to overcome this enduring trade-off between measurement richness and scalability (Brown et al., 2020). These models, pre-trained on vast datasets, possess remarkable capabilities for understanding and generating human language. Through prompt engineering—the process of designing instructions to guide model behavior—researchers can potentially leverage LLMs to perform complex analytical tasks (Liu et al., 2021; Xiao et al., 2023; Ornstein et al., 2025). However, merely demonstrating feasibility is insufficient; understanding how to reliably and

efficiently elicit high-quality annotations comparable to human experts remains a research challenge (Bojic et al., 2025), particularly for established, complex social science measures (Halterman and Keith, 2025).

This paper uses the Discourse Quality Index (DQI) (Steenbergen et al., 2003b) as a demanding case study to investigate these challenges. The DQI is a widely respected framework in political science for assessing the deliberative quality of political communication (Habermas, 1996; Bächtiger et al., 2018). Its multi-dimensional nature, reliance on contextual interpretation, and ordinal scoring scales make it notoriously labor-intensive to apply manually, limiting large-scale research. Automating the DQI represents a significant test for LLM capabilities in complex, theory-driven measurement.

While prior work suggests LLMs can perform qualitative coding tasks (Xiao et al., 2023; Omizo, 2024), this paper provides a systematic investigation into optimizing LLM-based DQI coding. Specifically, I make the following contributions: First, I evaluate the impact of In-Context Learning (ICL) example quantity on DQI coding accuracy for two modern LLMs (OpenAI’s gpt-4.1-nano and Google’s gemini-2.0-flash), systematically varying examples from zero-shot up to 50 to identify performance scaling patterns and optimal ranges (cf. Dong et al., 2024). Second, I conduct a comparative analysis of performance between these models, considering associated costs, to provide practical guidance on model selection for large-scale DQI annotation (cf. Bojic et al., 2025; Sun et al., 2024). Third, I perform a detailed error analysis on a high-performing configuration to understand the nature of residual errors and identify specific DQI dimensions or judgment types that remain challenging for current models (cf. Li, 2025). Fourth, I develop and open-source a modular, adaptable AI engineering pipeline that enables systematic experimentation with LLM prompting across various measurement tasks, providing a technical foundation that other researchers can build upon for their own qualitative measurement challenges.

This study demonstrates that carefully tuned prompting strategies, particularly leveraging an optimal number of few-shot examples, enable LLMs to perform complex qualitative

coding like the DQI at levels approaching expert human annotators. This significantly enhances the feasibility of large-scale deliberative quality research. However, the results presented here suggest that understanding model-specific performance and limitations through systematic evaluation and error analysis is vital for responsible application and identifying areas for future work, such as the exploration of more advanced reasoning prompts (Dunivin, 2024).

This paper proceeds as follows. Section 2 reviews related work in computational text analysis, LLMs for qualitative coding, and the DQI. Section 3 details my data, prompt design, models, and evaluation metrics. Section 4 presents the findings from the ICL scaling, model comparison, reliability analysis, and error analysis. Section 5 discusses the implications of these findings. Section 6 concludes.

2 Previous Work

This paper intersects with research on computational text analysis, the rapidly evolving capabilities of LLMs for complex tasks, and the specific challenge of measuring deliberative quality using the Discourse Quality Index (DQI).

2.1 Computational Methods for Measuring Textual Constructs

Early approaches to the quantitative analysis of text often relied on dictionary-based methods, counting words associated with specific concepts (Grimmer and Stewart, 2013; Tausczik and Pennebaker, 2010), though these methods face limitations in capturing context and subtlety (Ribeiro et al., 2016). Supervised machine learning techniques offered improvements by training models on human-labeled data to classify texts according to predefined categories (Rudkowsky et al., 2018; Card et al., 2015), but require significant labeling effort and may not generalize well. Unsupervised methods like topic modeling (Blei et al., 2003) and word embeddings (Mikolov et al., 2013) can uncover latent structures without labeled data (Garg

et al., 2018; Sagi and Deignan, 2013), but interpreting these structures and mapping them to specific theoretical constructs remains challenging. These methods generally struggle to replicate the fine-grained, multi-dimensional judgments typical of expert qualitative coding schemes like the DQI.

2.2 Large Language Models and In-Context Learning for Text Analysis

Recent advances in LLMs, powered by architectures like the Transformer (Vaswani et al., 2017), offer new possibilities for text analysis (Bommasani et al., 2021; Brown et al., 2020). These models learn rich representations of language during pre-training on vast corpora. A key feature is their ability to adapt to new tasks via prompt engineering, often without requiring task-specific fine-tuning (Liu et al., 2021). In-Context Learning (ICL), where the model learns from examples provided directly in the prompt, is particularly relevant for applying LLMs to specialized tasks with limited labeled data (Brown et al., 2020). Research suggests that performance often improves significantly with a small number of examples, followed by diminishing returns (Dong et al., 2024; Omizo, 2024). While other advanced techniques, such as Chain-of-Thought (CoT) reasoning, show promise for improving performance on complex tasks by prompting step-by-step logic (Wei et al., 2022; Kojima et al., 2022; Dunivin, 2024), this study focuses primarily on optimizing performance through the systematic variation of ICL examples.

2.3 Applying LLMs to Qualitative Coding and Political Text

There is a growing interest in applying LLMs to automate tasks traditionally performed through manual qualitative coding. Studies have demonstrated the potential of LLMs, particularly when guided by structured prompts mimicking codebooks (Xiao et al., 2023), to achieve reasonable agreement with human coders on various qualitative schemes (Omizo,

2024). Within political science and communication research, LLMs are being explored as tools for analyzing political texts (Ornstein et al., 2025), measuring concepts like ideology or sentiment, and potentially automating complex measurement tasks (Haltermann and Keith, 2025). Some work has begun applying LLMs to analyze debates or deliberation-like features, such as argument strength or interaction dynamics (Liang et al., 2024; Behrendt et al., 2024; Karanam et al., 2024), showing promise in capturing specific aspects of discourse.

2.4 The Discourse Quality Index: Measurement and Automation Efforts

The DQI, developed by Steenbergen et al. (2003b), remains a cornerstone for empirically assessing deliberative quality in political communication. Rooted in Habermasian discourse ethics, it provides a theoretically grounded, multi-dimensional framework designed to operationalize key deliberative ideals into measurable indicators.

The DQI conceptualizes deliberation through several core components, assessed typically at the level of the individual speech act. First, it considers basic **Participation**, evaluating whether speakers can engage freely without interruption (coded simply as impaired or normal). Second, it focuses heavily on the **Quality of Justification**, assessing both the *level* of reasoning (ranging from mere assertions to sophisticated chains of inference) and the *content* or nature of the reasons provided (distinguishing between appeals to narrow interests, the common good, or principles like Rawlsian justice). Third, the DQI measures **Respect** shown towards others in multiple forms: respect towards specific social groups, respect towards the *demands* or proposals of opponents, and respect towards *counterarguments* themselves (each assessed on separate ordinal scales reflecting increasing levels of acknowledgment or validation). Finally, it evaluates the **Constructiveness** of the political engagement, coding whether speakers engage purely positionally or make efforts towards compromise, consensus, or mediating proposals. These dimensions are summarized in Table 1. Each dimension is measured using carefully defined ordinal scales, requiring trained coders to make judgments

based on the textual content and its context.

The DQI framework has proven influential and has been widely applied across various contexts, including national parliaments (Wendler, 2013), televised debates (Rosset and Wüest, 2021), and online forums (Wisniewski, 2019), often requiring adaptation for specific settings (e.g., Bächtiger et al., 2009). Despite its utility and theoretical grounding, the detailed, multi-faceted nature of the DQI makes its manual application highly resource-intensive, limiting the scope and scale of empirical deliberation studies.

Consequently, automating the DQI has been a long-standing goal. Early attempts often relied on traditional machine learning techniques with feature engineering. While achieving some success, these approaches generally faced limitations in capturing the full complexity inherent in the DQI constructs, particularly for dimensions requiring deeper semantic understanding (Rincón et al., 2021; Fournier-Tombs and MacKenzie, 2021). While more recent work leverages the power of LLMs for related deliberative tasks or specific components (Liang et al., 2024; Behrendt et al., 2024; Karanam et al., 2024), a systematic evaluation of modern LLMs’ ability to replicate the original, complete DQI framework using optimized prompting strategies represents a research gap this study aims to fill.

3 Methodology

In this section, I detail the methodology used to evaluate the performance of large language models in automating the coding of the Discourse Quality Index (DQI).

3.1 Data

I use a validation dataset derived from Steenbergen et al. (2003b), consisting of 1044 parliamentary speeches from the 101st and 104th US Congress manually annotated by expert coders on the seven DQI dimensions (summarized in Table 1). The raw speech texts were sourced from the Congressional Record corpus (Gentzkow and Shapiro, 2017). Of the 1044

Dimension	Definition	Scoring Criteria
Participation	The extent to which participants are able to engage in the debate without being constrained.	0: Participation impaired, 1: Normal participation
Justification (Level)	The extent to which arguments are supported by reasons.	0: No justification, 1: Vague assertion, 2: General justification, 3: Specific justification
Justification (Content)	The quality of the justification provided.	0: Lack of facts, 1: Oversimplification, 2: Relevant examples, 3: Balanced consideration
Respect (Groups)	The extent to which participants show respect towards other groups.	0: No respect, 1: Formal respect, 2: Substantial respect
Respect (Demands)	The extent to which participants show respect towards the demands of other groups.	0: No respect, 1: Formal respect, 2: Substantial respect, 3: Acknowledges legitimacy
Respect (Counterarguments)	The extent to which participants show respect towards counterarguments.	0: No respect, 1: Formal respect, 2: Substantial respect, 3: Acknowledges validity, 4: Engages with counterargument
Constructive Politics	The extent to which participants engage in positive sum politics and try to reach compromise solutions.	0: Positional politics, 1: Alternative proposal, 2: Consensus appeal, 3: Mediating proposal

Table 1: DQI dimensions and scoring criteria

annotated speeches, 614 were complete, with annotations provided for each of the dimensions of the DQI. Following standard practice, these 614 speeches were split into a training set (80%, N=495) used solely for selecting In-Context Learning (ICL) examples, and a validation set (20%, N=119) used for evaluating model performance. The validation set remained unseen by the models during the example selection process.

3.2 Prompt Design and In-Context Learning

A prompt template was developed to provide the LLM with the necessary context and instructions for performing DQI coding. This template aimed to simulate the information

provided to a human coder, including a definition of the coder’s role (expert political scientist), a theoretical overview of the DQI, detailed coding criteria derived directly from the original DQI manual (Steenbergen et al., 2003b), a structured JSON output format specification (Figure 4), and the target speech text. A conceptual overview of the prompt structure is shown in Figure 1. For an example of a complete prompt, see the online Appendix.

```

You are an expert political scientist. Analyze the given
parliamentary speech and annotate it on the following dimensions
of the Discourse Quality Index (DQI).

Theoretical overview of the DQI:
{dqi-theory}

Annotation criteria for each dimension:
{dqi-criteria}

Structure annotations in the following format:
{output-format-json-schema}

Examples of annotated speeches:
{example_annotation-1}
...
{example_annotation-n} (*Varied from 0 to 50*)

Annotate the following speech:
{target-speech-text}

Final instructions:
{final-instructions}.

```

Figure 1: Conceptual overview of the prompt template structure provided to the Large Language Models.

Theoretical Foundation As specified in `dqi-theory`, the prompt provides an overview of the DQI framework and its theoretical foundations in the context of deliberative democracy. See Figure 2 for an excerpt of the theoretical foundation provided to the model. This formulation was taken directly from the appendix in Steenbergen et al. (2003a) that provides

the theoretical foundation of the DQI.

Dimension Definitions and Scoring Criteria The prompt details the dimensions and scoring criteria of the DQI, as outlined in `dqi-criteria`, providing descriptions and examples of the scoring levels for each dimension and guiding the model to consider evidence for each possible score before making a final decision. See 3 for an excerpt of the dimension definitions and scoring criteria provided to the model. As with the theoretical foundation, this information was taken directly from the appendix in Steenbergen et al. (2003a) that provides the dimension definitions and scoring criteria of the DQI.

Output Format As described in `output-format`, the prompt establishes a structured output format for the annotations, including fields for each dimension and sub-dimension, as well as additional fields for overall notes and summary if specified. See Figure 4 for the JSON¹ output format provided to the model.

Sample Speeches and Annotations The prompt includes a set of sample speeches with corresponding annotations, in `example-annotation-1` to `example-annotation-n`, with the goal of both establishing the expected level of detail and reasoning in the evidence/score justifications and providing a model for the annotator to follow when coding the target speech. To generate these examples, I select a random subset of speeches from the validation dataset and for each speech provide the complete speech text along with the corresponding DQI annotations. As I discuss in the next section, I experiment with different numbers of randomly selected examples to determine how responsive the model is to demonstration of the desired annotation behavior.

Target Speech The prompt presents the target speech to be annotated, in `target-speech`. The target speech is the main focus of the annotation task, and the model is directed to pro-

¹JSON (JavaScript Object Notation) is a lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.

The DQI attempts to measure political deliberation in a general, valid and reliable way (Steenbergen et al. 2003). It mainly draws on Habermasian discourse ethics, but also incorporates elements of other deliberative models. The unit of analysis of the DQI is a speech act, i.e. the discourse by a particular individual delivered at a particular point in a debate. For each speech, we distinguish between relevant and irrelevant parts, and only the relevant parts are coded. A relevant part is one that contains a demand, i.e. a proposal on what decision should or should not be made. Our emphasis on demands stems from the fact that they constitute the heart of the deliberation. That is, demands stipulate what ought to be and what ought not to be, and this normative character puts them at the center of discourse ethics. The DQI is composed of seven indicators. Despite the considerable complexity of parliamentary debates, we attempt to keep the coding categories relatively easy, so as to ensure a high level of reliability. The following is an elaboration of these indicators, followed by an overview of the indicators and their codes. We discuss seven the seven indicators under four headings.

1. Participation. Participation constitutes a fundamental precondition for deliberation. In parliamentary settings of western democracies, this type of basic participation can usually be seen as given for the elected representatives. Normal participation is only assumed to be impaired if a speaker is cut off by a formal decision, or if she or he feels explicitly disturbed in the case of a verbal interruption by other actors.

...

Figure 2: Excerpt of DQI Theoretical Foundation Provided To The Model

```
1. Participation
0: participation was impaired - speaker was cut off or
  explicitly disturbed
1: normal participation was possible

2. Justification
2.1 Level of Justification
0: no justification
1: inferior justification - reason given but not properly
  linked to demand
2: qualified justification - one complete linkage between
  demand and reason
3: sophisticated justification (broad) - multiple complete
  justifications
4: sophisticated justification (in depth) - multiple
  justifications with one embedded in complete inference
  chain

2.2 Content of Justification
0: explicit reference to group/constituency interests
1: neutral statement, no explicit references to group
  interests
2: explicit reference to common good (utilitarian/collective)
3: explicit reference to helping least advantaged (difference
  principle)

...
```

Figure 3: Excerpt of DQI Dimension Definitions and Scoring Criteria Provided To The Model

vide detailed annotations for each dimension of the DQI based on the content of the speech. I provide the first 10000 characters of the speech to ensure that the model has sufficient context to make informed judgments.

```
```json
{
 "participation": {"score": [0/1]},
 "justification": {
 "level": {"score": [0-4]},
 "content": {"score": [0-3]}
 },
 "respect": {
 "groups": {"score": [0-2]},
 "demand": {"score": [0-3]},
 "counterargument": {"score": [0-4]}
 },
 "constructive_politics": {"score": [0-3]}
}
```
```

Figure 4: Structured JSON output format required from the Large Language Models for DQI annotations.

In-Context Learning The core experimental manipulation focused on the effect of ICL example quantity. I generated prompt variants containing systematically varied numbers of examples: 0 (zero-shot), 5, 10, 15, 20, 25, 30, 35, 40, 45, and 50. Each example consisted of a speech text paired with its corresponding expert-coded JSON annotation, drawn randomly without replacement from the 800-speech training set for each annotation task performed on the validation set speeches. To mitigate the influence of idiosyncratic features arising from any single random draw of examples, a Monte Carlo simulation approach was employed. Specifically, for each model and for each ICL configuration containing more than zero examples (i.e., 5 through 50 examples), the entire process of randomly selecting examples and evaluating the model on the full validation set was repeated five independent times. The resulting performance metrics (e.g., MAE, Krippendorff’s Alpha) for each model at each

ICL level greater than zero were then averaged across these five draws to provide a more robust estimate of performance. This systematic variation, combined with the averaging over draws, allows for a clearer analysis of performance scaling as a function of the number of demonstrations provided to the models (cf. Dong et al., 2024).

3.3 Model Selection and Parameter Configuration

This study primarily evaluates and compares two contemporary LLMs: ‘gpt-4.1-nano’ (provided by OpenAI) and ‘gemini-2.0-flash’ (provided by Google). These models were selected because, as of this writing (April 2025), they represent the state-of-the-art in terms of capabilities-to-cost ratio for general-purpose language understanding. Both models are widely deployed in production applications and provide an optimal balance between performance and affordability, making them particularly suitable for resource-constrained research contexts that require processing substantial volumes of text.

The selection of these specific models was also driven by pragmatic considerations. The OpenAI and Google APIs offer high throughput and reliability, enabling efficient testing across multiple experimental conditions. Their well-documented interfaces and consistent performance characteristics facilitate systematic evaluation. While I acknowledge the concerns regarding closed-source models and the limitations this places on full scientific replicability, testing across the full spectrum of possible models (including open-source alternatives) falls outside the scope of this study. Instead, the primary research question focuses on whether affordable, modern LLMs can effectively perform complex qualitative coding tasks like the DQI with minimal fine-tuning, rather than identifying which specific architecture performs best.

For all API calls to these models, a temperature parameter of 0.7 was used to balance creativity and predictability in the generated annotations, and the maximum number of output tokens was set to 4000 to accommodate the structured JSON response. Other model parameters were left at their default API settings.

3.4 Evaluation Metrics

Model performance is primarily evaluated by comparing the LLM’s predicted DQI scores ($\hat{y}$) against the expert ground truth scores (y) from the validation set (N=119 speeches). Two key metrics are employed.

First, Mean Absolute Error (MAE) is calculated, averaged across all M=7 DQI dimensions. MAE provides an intuitive measure of the average magnitude of error on the original DQI scales:

$$\text{MAE} = \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M |\hat{y}_{ij} - y_{ij}|$$

Lower MAE values indicate better performance. Considering typical human inter-rater reliability for DQI (e.g., Neblo et al., 2018; Steiner et al., 2004, $\alpha = 0.7 - 0.8$), an average MAE below 1.0 suggests performance approaching human consistency levels.²

Second, Krippendorff’s Alpha (α) (Krippendorff, 2018) is calculated to assess inter-rater reliability between the LLM and the human gold standard. Alpha was computed separately for each DQI dimension using the ordinal level of measurement setting in the `krippendorff` Python package. This accounts for the magnitude of disagreements on the ordinal scales and allows for direct comparison to reliability benchmarks commonly reported in DQI research and LLM annotation studies (cf. Bojic et al., 2025). Higher Alpha values (approaching 1.0) indicate better agreement than expected by chance.³

²While MAE measures average error magnitude and Krippendorff’s Alpha measures chance-corrected agreement, they provide complementary views on performance. Human coders achieving acceptable reliability (e.g., $\alpha = 0.7 - 0.8$) on ordinal scales like the DQI (typically ranging 0-3 or 0-4) inevitably exhibit some disagreement. An MAE represents the average absolute difference between two sets of ratings. If an LLM achieves an MAE substantially below 1.0 compared to the human gold standard, it implies its average deviation is likely within the typical range of point differences observed between two reasonably consistent human coders. Thus, $\text{MAE} < 1.0$ serves as a useful, albeit heuristic, benchmark suggesting the LLM’s error magnitude is comparable to expected human variation.

³For two coders, Krippendorff’s α is

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = \frac{1}{n} \sum_{i < j} n_{ij} \delta_{ij}, \quad D_e = \frac{1}{n(n-1)} \sum_c \nu_c(n - \nu_c) \bar{\delta}_c$$

where n is the number of items, n_{ij} the frequency of the (coder 1, coder 2) category pair (i, j) , ν_c the marginal frequency of category c , and δ_{ij} the ordinal-level distance between categories i and j . For ordinal data δ_{ij} is the squared difference of cumulative relative frequencies: $\delta_{ij} = (F_i - F_j)^2$, with $F_k = \sum_{m \leq k} \nu_m / n$.

3.5 Pipeline Architecture

To systematically evaluate the performance of LLMs on the DQI coding task across different ICL configurations, I developed a modular pipeline architecture that automates the end-to-end experimentation process. The pipeline consists of five key stages: (1) data preparation, (2) prompt generation, (3) LLM API interaction, (4) response parsing, and (5) evaluation and visualization. Each stage is implemented as a separate module in Python, enabling flexible configuration and independent execution. This architecture facilitates rigorous experimental control while efficiently managing the complexity of testing multiple models, ICL configurations, and random initializations.

Beyond its use in this specific study, this pipeline architecture represents a significant contribution to the broader research community. Designed to be adaptable and extensible, it provides researchers with a framework for conducting their own systematic prompting experiments across various measurement tasks in the social sciences. The pipeline’s modular design allows components to be replaced or modified to accommodate different data sources, prompting strategies, models, and evaluation metrics, making it suitable for diverse research applications beyond DQI coding.

The pipeline is orchestrated using a Makefile system with a centralized configuration file that defines parameters for each model, ICL setting, and experimental condition. This approach ensures reproducibility by maintaining consistent experimental protocols and recording all generated artifacts. The pipeline handles practical challenges such as rate limiting for API calls, parallel processing of requests, and validation of LLM outputs.⁴

D_o is the observed disagreement, D_e the disagreement expected by chance; $\alpha = 1$ when $D_o = 0$ and $\alpha = 0$ when $D_o = D_e$.

⁴A complete implementation is available for researchers to use, adapt, and extend at <https://github.com/mbosley/mbosley.github.io>.

3.6 Error Analysis Approach

To gain deeper insights into model limitations, a detailed error analysis was conducted on the predictions generated by a high-performing configuration (`gemini-2.0-flash` with 50 ICL examples, initialization 0). This analysis involved several steps.

First, I Examining MAE and Alpha scores per DQI dimension to identify which aspects of deliberative quality were most challenging for the target LLM (`gemini-2.0-flash`) to code reliably; (2) generating confusion matrices for each dimension to visualize patterns in score disagreements (e.g., frequent confusion between adjacent scores).

Second, I perform a systematic qualitative review, assisted by a separate, powerful LLM (Gemini 2.5 Pro). Specifically, Gemini 2.5 Pro was employed to conduct an initial, detailed comparison between the `gemini-2.0-flash` predictions and the expert human gold standard codes for a sample of 30 validation set speeches. For each speech in this sample, Gemini 2.5 Pro identified the dimensions of disagreement and provided a preliminary qualitative analysis of the potential reasons for the discrepancies. The full output of this LLM-assisted comparison is documented in the online appendix.

Third, I synthesized the findings from this systematic, LLM-assisted review, interpreting the detailed comparisons from the appendix to identify recurring error patterns, challenging linguistic features, and key areas where the target model (`gemini-2.0-flash`) struggled (cf. Li, 2025). The summary of these synthesized error patterns is presented in the main text. This collaborative approach leverages the systematic analysis capabilities of a powerful LLM for the detailed comparison task, while retaining human oversight and interpretation for identifying the broader themes and implications of the errors.

4 Results

This section presents the results of the experiments evaluating the performance of ‘gpt-4.1-nano’ and ‘gemini-2.0-flash’ on the DQI coding task across different numbers of In-Context

Learning (ICL) examples. Performance is assessed primarily using Mean Absolute Error (MAE) and Krippendorff’s Alpha (ordinal).

4.1 Impact of In-Context Learning Examples on Overall Performance

Figure 5 illustrates the average MAE across all seven DQI dimensions as a function of the number of ICL examples provided in the prompt. Both models demonstrate a substantial improvement over their zero-shot performance when provided with even a small number of examples. The ‘gemini-2.0-flash’ model starts with a lower zero-shot MAE (approximately 0.61) compared to ‘gpt-4.1-nano’ (approximately 0.84).

The most significant reduction in error for both models occurs between 0 and 10-15 examples. By 10 examples, both models converge to a similar MAE level of approximately 0.44. Beyond this point, their performance characteristics diverge slightly. The ‘gemini-2.0-flash’ model’s performance stabilizes, achieving its lowest average MAE around 30-50 examples (approximately 0.40). In contrast, the ‘gpt-4.1-nano’ model exhibits more fluctuation after 10 examples, with MAE varying between roughly 0.44 and 0.54, generally settling at a slightly higher error level than ‘gemini-2.0-flash’. These results indicate diminishing returns for additional ICL examples beyond approximately 15-20 examples, suggesting that providing a moderate number of examples is sufficient to capture most of the achievable performance gains for this task, consistent with general findings on ICL scaling (Dong et al., 2024; Omizo, 2024).

4.2 Dimension-Specific Performance and Model Comparison

Performance varies considerably across the individual DQI dimensions, as shown in Figure 6. Some dimensions proved much easier for the models to learn than others. "Participation" was coded with near-perfect accuracy (MAE close to 0) by both models almost immediately

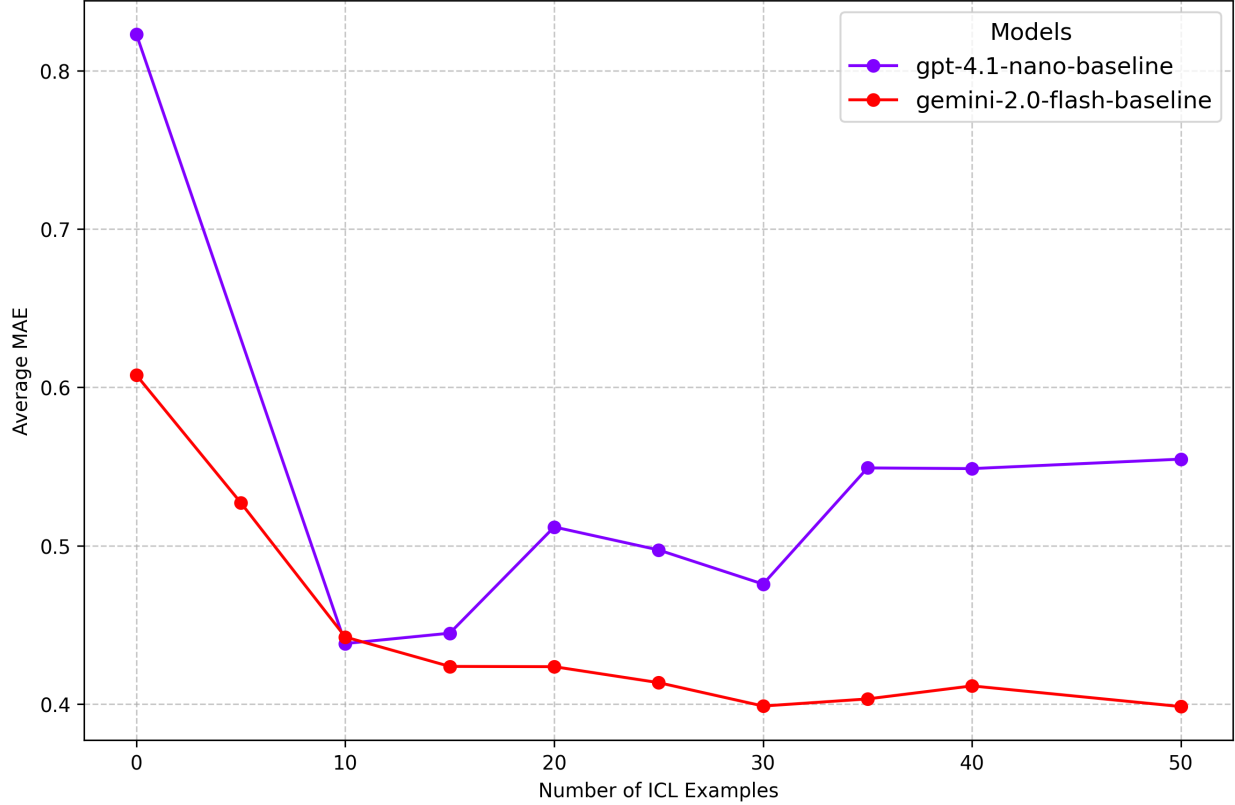


Figure 5: Overall Mean Absolute Error (MAE) across all DQI dimensions versus the number of In-Context Learning (ICL) examples provided in the prompt, shown for ‘gpt-4.1-nano’ (Purple) and ‘gemini-2.0-flash’ (Red). Points represent average MAE over N=119 validation speeches, averaged over 5 draws for ICL > 0.

upon inclusion of any ICL examples. "Constructive Politics" also showed strong performance, with both models achieving low MAE (often below 0.2) after approximately 15 examples; ‘gemini-2.0-flash’ consistently achieved slightly lower error on this dimension in the 15-50 example range.

Conversely, certain dimensions remained challenging. "Justification-Level" exhibited persistently high MAE (above 0.8 for ‘gpt-4.1-nano’ beyond 15 examples, and above 0.6 for ‘gemini-2.0-flash’). Notably, ‘gemini-2.0-flash’ consistently outperformed ‘gpt-4.1-nano’ on this dimension, and the latter model’s performance actually degraded slightly as more examples were added beyond 15. "Respect-Counterarg" also proved difficult, with MAE remaining relatively high (around 0.8-1.0) for both models, showing only modest improvement

with ICL.

Performance on intermediate dimensions varied. For "Respect-Group", 'gemini-2.0-flash' achieved substantially lower MAE (around 0.1-0.15) than 'gpt-4.1-nano' after 10 examples. On "Respect-Demand", both models improved significantly with ICL, reaching comparable performance levels with MAE plateauing around 0.4-0.5. For "Justification-Content", both models reduced MAE considerably with examples but stabilized at a moderate error level (around 0.5-0.6).

Across dimensions, 'gemini-2.0-flash' generally displayed more stable error rates after the initial rapid improvement phase (0-15 examples). It demonstrated a clear advantage on "Justification-Level" and "Respect-Group" compared to 'gpt-4.1-nano' in the few-shot setting.

4.3 Reliability Analysis

To assess the reliability of the LLM annotations against the human gold standard, Krippendorff's Alpha (ordinal) was calculated for each dimension. Figure 7 presents these results across different ICL levels for both models. The Alpha scores generally mirror the MAE findings, with dimensions exhibiting low MAE achieving higher reliability scores.

Several dimensions reached levels of reliability often considered "Substantial" (α between 0.67 and 0.80) or "Excellent" ($\alpha > 0.80$) in social science research (cf. Bojic et al., 2025; Krippendorff, 2018) when sufficient ICL examples (typically 10-15 or more) were provided. Specifically, "Participation" consistently achieved near-perfect reliability. "Respect-Group", "Respect-Demand", and "Constructive Politics" frequently reached substantial or excellent agreement for both models, with 'gemini-2.0-flash' often achieving higher scores or reaching thresholds with fewer examples, notably hitting excellent reliability on "Respect-Group".

In contrast, the dimensions identified as challenging based on MAE also showed low reliability. "Justification-Level" and "Respect-Counterarg" typically exhibited Alpha scores in the "Poor" ($\alpha < 0.20$) to "Fair" ($0.20 \leq \alpha < 0.40$) ranges, indicating that the models'

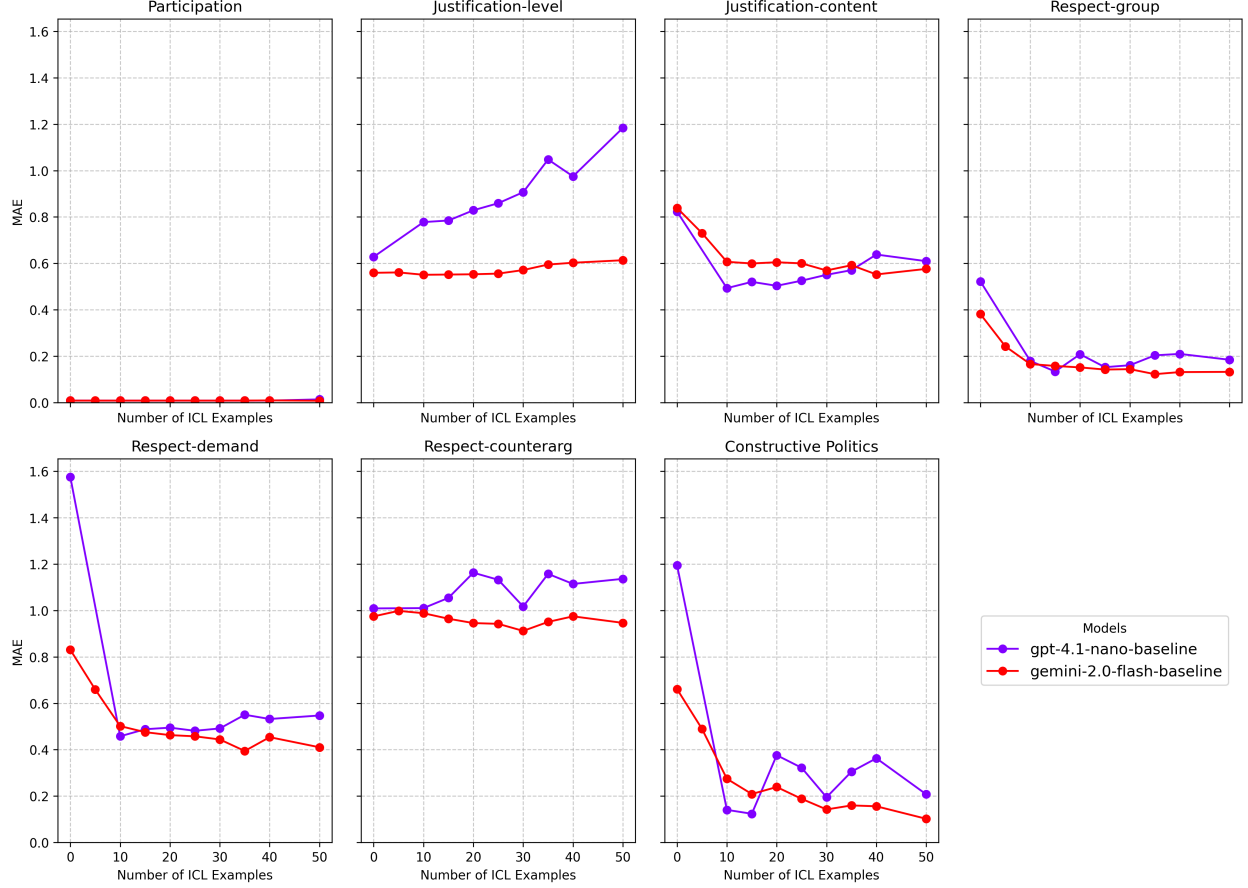


Figure 6: Mean Absolute Error (MAE) versus the number of In-Context Learning (ICL) examples, faceted by each of the seven DQI dimensions. Lines show performance trends for ‘gpt-4.1-nano’ (Purple) and ‘gemini-2.0-flash’ (Red) on each specific dimension, averaged over 5 draws for ICL > 0.

ratings on these dimensions had agreement levels only slightly better, or sometimes worse, than chance compared to the human coders. "Justification-Content" generally achieved "Moderate" reliability ($0.40 \leq \alpha < 0.67$).

The positive effect of ICL examples on reliability is evident across most dimensions, with Alpha scores generally increasing and stabilizing after 10-15 examples, mirroring the MAE stabilization point. Table 2 summarizes the reliability and MAE scores for a high-performing configuration. Overall, while the models demonstrate the capacity for reliable coding on several DQI dimensions comparable to human standards, significant reliability challenges persist for others.

Reliability by Dimension Across Models

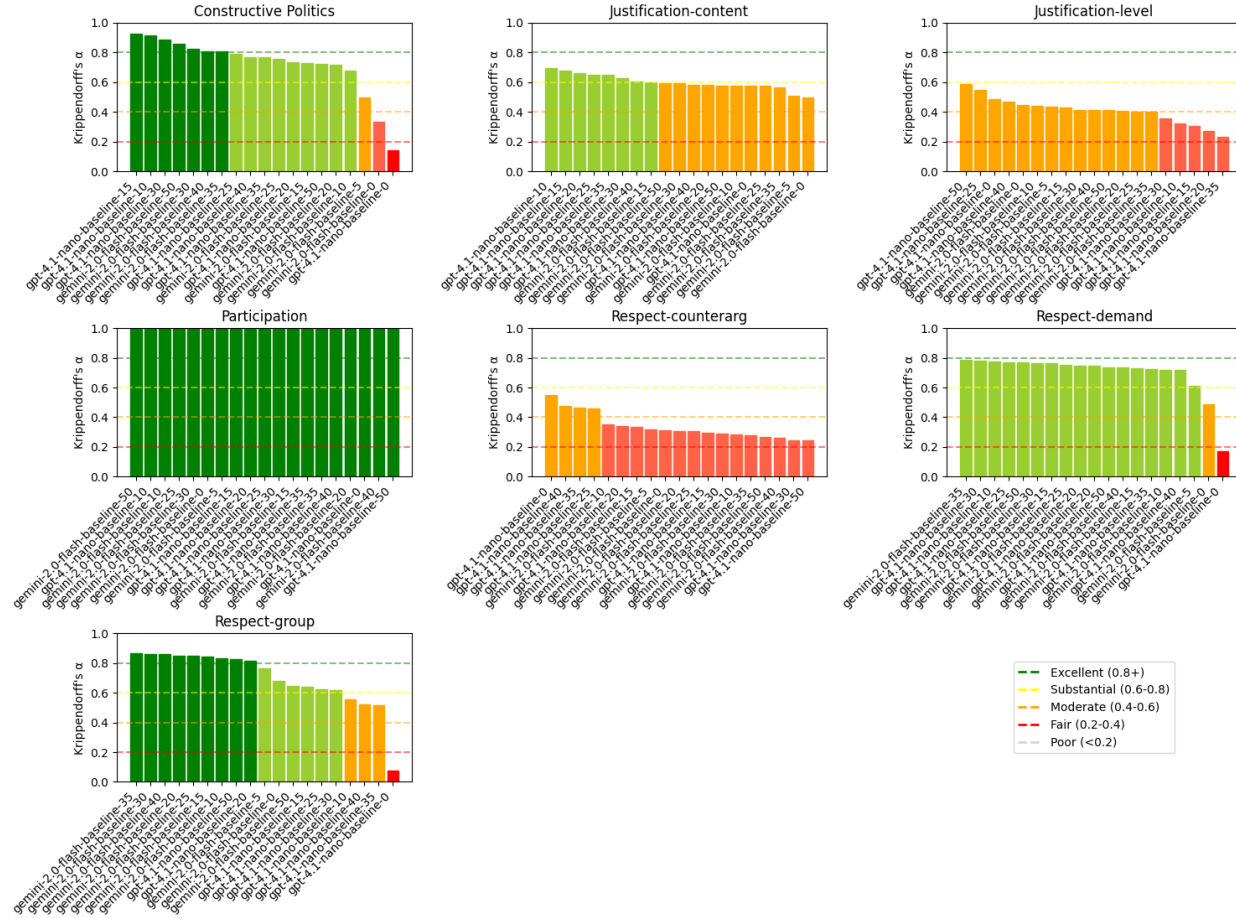


Figure 7: Krippendorff's Alpha (Ordinal) by DQI Dimension for 'gpt-4.1-nano' and 'gemini-2.0-flash' across different numbers of ICL examples (0-50). Bars represent Alpha values, averaged over 5 draws for ICL > 0. Dashed lines indicate conventional reliability benchmarks.

| DQI Dimension | Krippendorff’s α (ordinal) | MAE |
|-----------------------|-----------------------------------|-------|
| Participation | 0.9996 | 0.008 |
| Justification–Level | 0.4143 | 0.614 |
| Justification–Content | 0.5936 | 0.576 |
| Respect–Group | 0.6446 | 0.132 |
| Respect–Demand | 0.7683 | 0.410 |
| Respect–Counterarg | 0.2652 | 0.947 |
| Constructive Politics | 0.8597 | 0.102 |
| Average | 0.6493 | 0.398 |

Table 2: Performance of `gemini-2.0-flash` (50 ICL examples) on the validation set. Values are averaged over the five random-seed initializations; lower MAE and higher α indicate better agreement with expert labels.

4.4 Cost Comparison

Beyond accuracy and reliability, the practical feasibility of using these models at scale depends on their computational cost. Figure 8 presents the total cost incurred to process the entire validation set (N=119 speeches) for each model and ICL configuration.

As expected, the cost increases substantially with the number of ICL examples for both models. This is because adding examples directly increases the length of the prompt sent to the model, thereby increasing the number of input tokens processed per API call. The cost ranges from approximately \$0.03 for zero-shot inference to around \$0.50 for inference with 50 examples, representing a roughly 15-17 fold increase across this range.

Comparing the two models reveals a consistent pattern. While ‘`gpt-4.1-nano`’ is marginally cheaper in the zero-shot setting (\$0.032 vs \$0.034 for ‘`gemini-2.0-flash`’), it becomes progressively more expensive than ‘`gemini-2.0-flash`’ as the number of ICL examples increases. For instance, at 50 examples, processing the validation set cost \$0.525 with ‘`gpt-4.1-nano`’ compared to \$0.487 with ‘`gemini-2.0-flash`’. This trend holds across all tested few-shot levels (5 through 50 examples).

The performance analyses (Figures 5 and 6) showed that performance gains diminish sig-

nificantly beyond 15-20 examples, with ‘gemini-2.0-flash’ achieving near-optimal and stable performance around 30 examples (MAE = 0.40). However, the cost continues to rise steeply beyond this point. Using 50 examples instead of 30 increases the cost for ‘gemini-2.0-flash’ by nearly 90% (from \$0.257 to \$0.487) for virtually no improvement in average MAE. Similarly, for ‘gpt-4.1-nano’, increasing examples from its performance plateau around 10-15 examples up to 50 leads to a large cost increase (from \$0.17 to \$0.525) without a clear, sustained improvement in average performance.

These findings highlight a clear cost-performance trade-off. While adding more examples up to a point is essential for accuracy, exceeding the saturation point (15-30 examples for this task) dramatically increases costs with little to no corresponding benefit in overall coding quality. Furthermore, considering both performance and cost, gemini-2.0-flash appears generally more cost-effective in the few-shot regime for this specific DQI coding task, offering slightly better and more stable accuracy at a lower computational expense compared to ‘gpt-4.1-nano’ when using 5 or more examples.

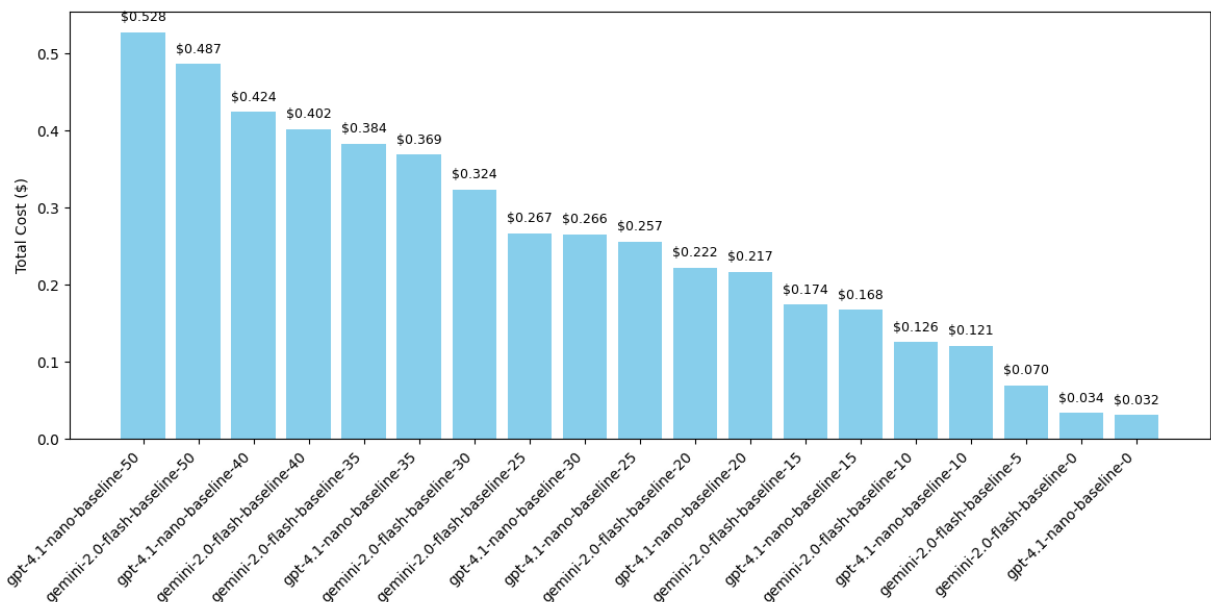


Figure 8: Total Cost (USD) to process the validation set (N=119 speeches) for gpt-4.1-nano and gemini-2.0-flash across different numbers of ICL examples (0-50).

4.5 Error Analysis

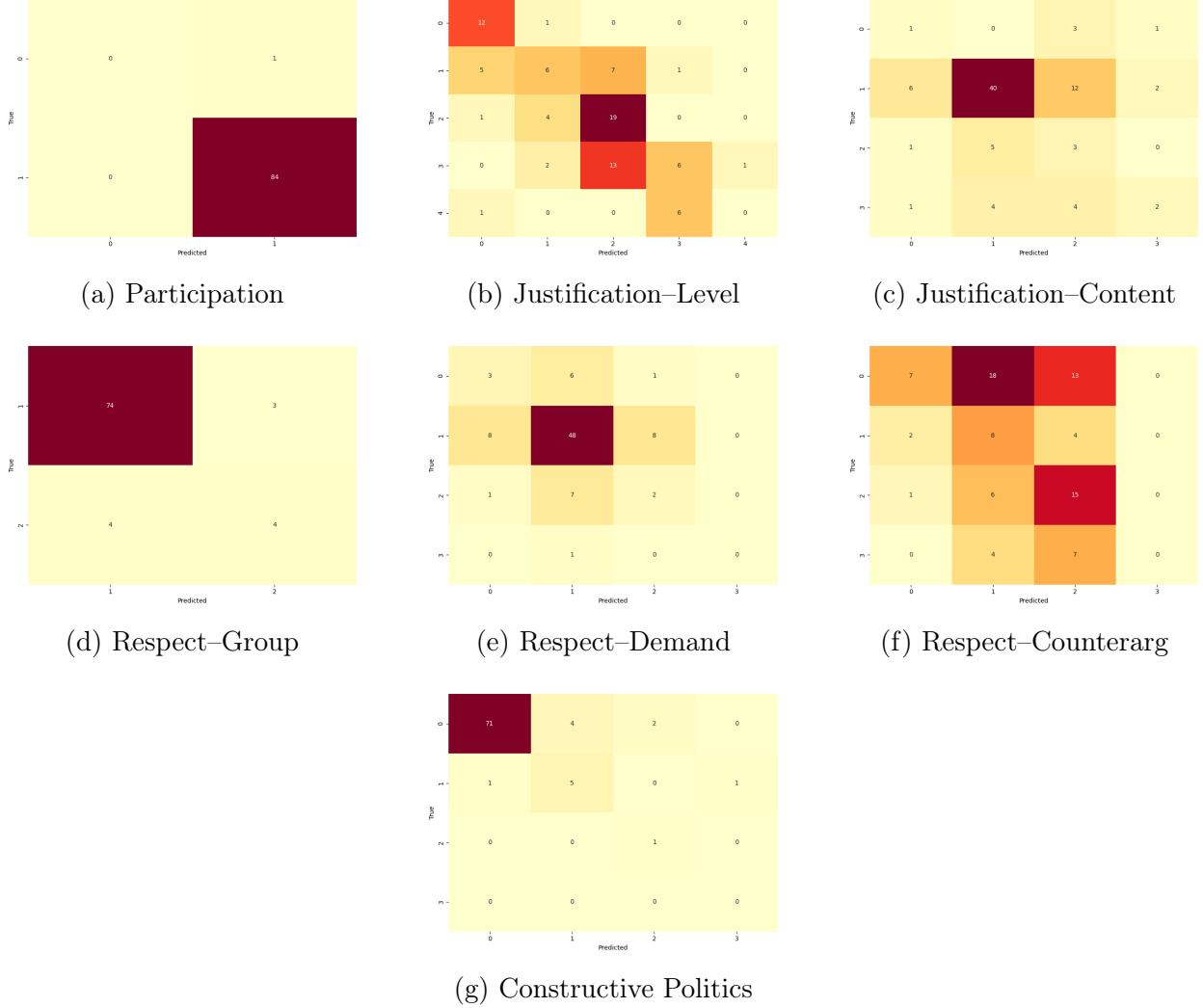


Figure 9: Confusion matrices for `gemini-2.0-flash` (50 ICL examples, init 0) on the validation set ($N = 119$). Rows denote expert labels; columns denote model predictions; darker cells indicate more items.

While the aggregate performance metrics demonstrate considerable success, particularly with sufficient ICL examples, a detailed qualitative analysis focusing on the `gemini-2.0-flash` model (50 ICL examples) provides deeper insights into the model’s coding behavior and reveals specific, recurring patterns that explain the observed performance variations across dimensions. Examination of confusion matrices (Figure 9) alongside qualitative review of individual speech discrepancies (drawing from the sample analyzed in the online Appendix)

highlights these tendencies.

The analysis confirms the model’s strengths on dimensions with clear, often surface-level cues. As indicated by the near-perfect confusion matrix for "Participation" (Figure 9a), performance was nearly flawless, likely because the model can easily identify the absence of explicit interruptions mentioned in the transcript. Similarly, dimensions achieving high reliability like "Constructive Politics" (Figure 9g) and "Respect-Group" (Figure 9d) show strong diagonal concentrations in their matrices for the most frequent categories. The model reliably identifies "Positional Politics" (Level 0) for Constructive Politics (71/77 cases in the sample analyzed) and "Neutral/Implicit" respect (Level 1) for Respect-Group (74/77 cases). However, qualitative review revealed minor systematic tendencies even here. For Constructive Politics, there was a slight over-ascription of compromise based on procedural or detailed legislative outlining. For instance, in one speech (Speech 21 by Mr. Biden), a lengthy section detailing and comparing the features of the Democratic crime bill versus the President’s crime bill, while clearly advocating for the former, was coded by the model as a "Mediating Proposal" (Level 2), likely triggered by the detailed outlining of alternative policy frameworks, whereas human coders saw it as standard positional advocacy (Level 0). For Respect-Group, the model sometimes under-detected positive respect unless strong positive sentiment words were present. In Speech 15, Mr. Fazio states: “Our approval of H.R. 2 will send a strong signal that we value these workers and their role in American society.” This was coded as Level 1 (Neutral) by the model, whereas human coders recognized the explicit positive valuation of "workers" as Level 2 respect.

For dimensions showing moderate overall performance, specific error patterns explain the observed confusion. In "Justification-Content" (Figure 9c), while the model reliably identifies the frequent "Neutral" category (Level 1, 40/58 correct in sample), its difficulty differentiating value-based justifications (Levels 0, 2, 3) often stems from a heuristic focus on entity keywords. For example, in Speech 1, Mr. Dreier quotes a letter arguing against the minimum wage:

This line of thinking runs counter to the basic principles of a free society. Government edicts do not make wages rise; they rearrange and redistribute existing wealth. Wages rise in response to the creation of new wealth through greater productivity. Government cannot create or produce anything. Mr. Speaker, we should in fact reject this. We should improve the standards of those at the lower end of the economic spectrum by decreasing the tax and regulatory burden imposed on the private sector.

Human coders identified the appeal to "basic principles of a free society" and "greater productivity" as Level 3 (universal principles/common good), but the model focused on the mention of the "private sector," coding it as Level 1 (self/group interest). Similarly, for "Respect-Demand" (Figure 9e), where the common "Neutral/Implicit" respect category (Level 1) is relatively well-identified (48/64 correct), errors often involve misclassifying both disrespect (Level 0) and explicit respect (Level 2) as Level 1. This arises from the model's tendency to under-detect subtle respect cues common in legislative discourse. Standard procedural statements of support, such as Mr. Alexander's in Speech 3: "Mr. Chairman, I rise in support of the gentlemen's amendment." which human coders typically register as explicit respect (Level 2), were frequently coded by the model as merely neutral (Level 1), likely because it lacks overt positive sentiment beyond the procedural function.

The dimensions identified as most difficult—"Justification-Level" (Figure 9b) and "Respect-Counterarg" (Figure 9f)—exhibit the most significant off-diagonal spread in their confusion matrices, and the qualitative analysis pinpoints the reasons. For "Justification-Level", where the model often scores one level lower than humans (note the significant number of true Level 3 cases predicted as Level 2), the bias appears driven by an over-reliance on explicit discourse markers. The model seems to look for words like "because" or "therefore" and struggles when reasoning is implied or embedded within dense examples or comparisons without clear argumentative signposting. Consequently, it frequently downgrades justifications that humans perceive as well-supported. In Speech 23, Mr. Gekas provides a detailed comparison:

For instance, the bill that is before us says that the death penalty by and large must be relegated to only those killings which come about with an intent to kill, but we say... when an actor... acts with reckless disregard... like a driveby killing... we believe that that individual should be just as subject to the possibility of the death penalty... That is an outstanding difference... Here we are again, but here is a significant difference that proves our point by the exception that the majority put in their bill... they put in a provision that says a driveby killing... What does that mean? It means they drew it so narrowly... that if the individual... steps out of that car... he would not be subject... So the answer is to impose our standard...

Human coders identified this detailed, multi-point critique and proposal as sophisticated justification (Level 3), but the model coded it as only qualified justification (Level 2), perhaps missing the linkages between the examples and the broader argument due to the lack of simple connective phrases.

Similarly, the widespread confusion in "Respect-Counterarg" reflects challenges beyond simple adjacent-category errors. The model struggles to track prior arguments and evaluate the nature of speaker engagement with them. Its tendency to equate lack of insults with neutrality (Level 1) and overlook pragmatic politeness leads to frequent under-scoring of respect. Furthermore, it has difficulty interpreting common rhetorical structures like acknowledging an opponent's position respectfully before refuting it. In Speech 7, Mr. Bartlett states:

Mr. Chairman, I rise in very strong opposition to the amendment offered by the gentleman from Virginia [Mr. OLIN]. I have a great deal of respect for the gentleman from Virginia (Mr. OLIN). I have a great deal of respect for those who would urge this amendment... In my Opinion, though, this amendment would In fact be harmful...

Human coders identified the explicit statements of respect as valuing the counterargument (Level 3), despite the strong opposition. The model, however, coded this as degrading the

counterargument (Level 1), seemingly unable to reconcile the explicit respect with the subsequent criticism, and perhaps interpreting the standard parliamentary courtesy as insincere or overridden by the opposition. Identifying these higher levels of respectful engagement (Level 3: explicitly valuing) proved particularly challenging for the model, with no instances correctly classified as Level 3 in the analyzed sample where the human code was 3.

In summary, the error analysis highlights that while the LLM demonstrates proficiency, particularly on dimensions with clear lexical cues, its coding strategy differs from human coders on more interpretively complex aspects. It appears to rely more on shallower discourse parsing, explicit linguistic tokens, and heuristics (like entity mentions triggering content codes or procedural language indicating support/opposition), while struggling with implied reasoning, pragmatic politeness norms distinct from overt sentiment, and tracking inter-speech argumentative relationships. These systematic differences, rather than purely random errors, explain the persistent challenges on dimensions like Justification-Level and Respect-Counterarg.

5 Discussion

The results presented in this study offer insights into the capabilities and optimization of Large Language Models for automating complex, theory-driven qualitative coding in political science, and underscore both the potential of LLMs to transform large-scale textual analysis and the importance of carefully evaluating these models' outputs.

A central finding concerns the efficacy and scaling behavior of In-Context Learning. The dramatic improvement in performance when transitioning from zero-shot to few-shot prompting confirms the power of providing demonstrations for complex, rule-based tasks like DQI coding. More importantly, the observed saturation effect, with performance plateauing around 15-30 examples, aligns with broader trends identified in ICL research (Dong et al., 2024) and specific qualitative coding applications (Omizo, 2024). This saturation point has

profound practical implications: it suggests that achieving high levels of accuracy does not necessitate the creation of vast, task-specific training datasets often required by traditional supervised learning. Instead, a carefully curated set of a few dozen high-quality examples can effectively guide modern LLMs to internalize complex coding criteria, significantly lowering the barrier to entry for applying these powerful models to bespoke social science measurement tasks.

The comparison between ‘gpt-4.1-nano’ and ‘gemini-2.0-flash’, while specific to these models and this task, highlights the interplay between model architecture and prompt engineering. Although ‘gemini-2.0-flash’ generally exhibited slightly more stable and often slightly better performance, particularly on challenging dimensions like "Justification-Level", both models achieved strong results with sufficient ICL examples. This suggests that while inherent model differences exist and may confer advantages on certain sub-tasks, effective prompt engineering—particularly the provision of relevant examples—can substantially shape and improve performance across different capable models. Researchers are therefore not solely dependent on accessing the single "best" model but can potentially achieve high performance with various LLMs through careful prompt design.

Perhaps the most significant outcome is the demonstration that LLMs can achieve levels of inter-rater reliability comparable to human standards on several core DQI dimensions. Reaching Krippendorff’s Alpha scores above the conventional threshold for substantial agreement ($\alpha > 0.67$) for dimensions like "Participation," "Respect-Group," "Respect-Demand," and "Constructive Politics" (cf. Bojic et al., 2025) is noteworthy. It indicates that, for these facets of deliberation, the LLM can replicate expert human judgments with a consistency deemed acceptable for rigorous social science research. This capability marks a clear advance over earlier automation attempts using classical machine learning, which often struggled to capture the required interpretive depth, hitting performance ceilings well below human reliability levels (Rincón et al., 2021; Fournier-Tombs and MacKenzie, 2021).

However, this study also clearly maps the current frontiers and limitations. The persistent

difficulties encountered with "Justification-Level" and "Respect-Counterarg" are instructive. The detailed error analysis, including the qualitative review and confusion matrix patterns, suggests these failures stem from the models' reliance on more superficial linguistic cues and heuristics, rather than a deeper semantic or pragmatic understanding comparable to human experts. For instance, the model's struggle with implied reasoning, its tendency to underweight pragmatic politeness cues common in legislative discourse, and its heuristic classification of justification content based on entity mentions point to specific gaps. These are not merely random errors but indicate systematic differences in how the LLM processes information compared to a trained human coder applying the DQI framework. While the models are often "close" (as indicated by many errors being between adjacent categories), reliably making these fine-grained distinctions on the most complex dimensions remains a challenge for current architectures guided solely by ICL.

These findings carry substantial implications for empirical research on deliberation and for computational social science methodology. The ability to reliably automate coding for several DQI dimensions dramatically expands the scope of possible research. Large-N analyses of parliamentary debates across time or institutions, comparative studies of online deliberation platforms, or dynamic tracking of discourse quality become significantly more feasible. This potential to analyze previously intractable volumes of text could unlock new insights into the conditions and consequences of deliberative politics. Yet, the variable performance across dimensions suggests that the annotations of generative LLMs shouldn't be trust implicitly. Researchers must be transparent about which DQI dimensions their automated analysis captures reliably. Drawing strong conclusions based solely on dimensions where LLM performance is currently weak (like Justification-Level) would be inappropriate. Instead, studies might focus analyses on the reliably coded dimensions, employ automated coding as a first pass to identify segments for closer human analysis, or integrate LLM outputs with human validation, particularly for the most challenging constructs.

The limitations of this study—its focus on a specific dataset from the US Congress, the

evaluation of only two LLMs, and the use of one specific prompt implementation—naturally suggest avenues for future research. Testing generalizability across different political contexts, languages, and text genres is paramount. Continued evaluation of newer LLM architectures will be essential as the technology evolves. Future work should focus on targeted strategies to improve performance on the identified challenging dimensions.

6 Conclusion

This paper systematically evaluated the use of modern Large Language Models (OpenAI’s ‘gpt-4.1-nano’ and Google’s ‘gemini-2.0-flash’), guided by In-Context Learning, to automate the coding of the Discourse Quality Index in parliamentary speech. The core contributions lie in demonstrating not only the feasibility but also the optimization parameters for applying LLMs to this complex, multi-dimensional framework, and in developing a modular AI engineering pipeline that can be adapted for various measurement tasks across the social sciences.

The findings confirm that LLMs can achieve high fidelity in coding several DQI dimensions, reaching levels of reliability comparable to expert human annotators when provided with a moderate number of ICL examples (typically 15-30). Performance rapidly improves from zero-shot to few-shot settings, with diminishing returns beyond this optimal range. While model choice influences performance on specific dimensions, effective ICL prompting enables strong results from different capable models. However, the study also highlights current limitations, with dimensions requiring deep assessment of justification quality and counterargument engagement remaining challenging for the models, as evidenced by lower reliability scores and specific error patterns.

This research supports the argument that prompt-engineered LLMs offer a powerful tool for significantly enhancing the scale and efficiency of complex qualitative measurement in the social sciences. By carefully selecting models and optimizing prompting strategies, par-

ticularly the number of ICL examples, researchers can automate substantial portions of labor-intensive coding tasks like the DQI, enabling new avenues for large-scale empirical inquiry into phenomena like deliberative democracy. The open-source pipeline architecture developed for this study provides a practical framework for researchers to conduct their own systematic prompting experiments, further advancing the application of AI to measurement challenges in social science research.

Future work should focus on addressing the identified limitations. A promising direction involves exploring advanced prompting techniques, such as Chain-of-Thought reasoning (Dunivin, 2024; Wei et al., 2022), specifically targeted at improving performance on the dimensions where models currently struggle (Justification-Level, Respect-Counterarg). Additionally, investigating whether more computationally intensive "thinking-optimized" models, such as OpenAI's O-series, might yield better results on these complex dimensions represents a particularly intriguing avenue. These models, which trade efficiency for deeper reasoning capabilities, might better handle the contextual interpretation required for dimensions like Justification-Level that exhibit persistent challenges with current models, potentially echoing the inference-time scaling results observed in computer science research (Brown et al., 2024). The pipeline architecture developed for this study can be readily adapted to investigate such questions with minimal modifications.

Beyond model exploration, investigating the potential benefits of model fine-tuning (Halterman and Keith, 2025) or using ensemble methods combining multiple model judgments (Behrendt et al., 2024) also warrants attention. Expanding the evaluation to more diverse datasets, institutional contexts, and languages is crucial for assessing the generalizability of these findings. Ultimately, developing robust human-in-the-loop systems, where LLMs handle high-reliability coding and flag difficult cases for human review, may offer the most effective balance between scale, efficiency, and accuracy for deploying these powerful tools in social science research.

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